

INTRODUCTION

This exploratory data analysis (EDA) was conducted to examine patterns, trends, and relationships within an e-commerce sales dataset. The analysis involved summarising key variables, visualising business metrics, and identifying insights related to product performance, customer behaviour, payment methods, referral sources, order statuses, and monthly sales trends.

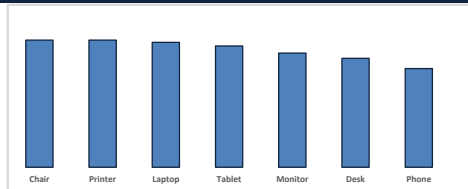
STATISTICAL OVERVIEW

Statistics	Quantity	UnitPrice	ItemInCart	TotalPrice
Count	1200	1200	1200	1200
Mean	2.95	356.41	4.89	1053.97
Median	3	364.21	5	823.62
Minimum	1	11.39	1	11.39
Maximum	5	699.93	10	3456.40

The statistical overview indicates that the dataset contains 1,200 observations across all numerical variables. On average, customers purchase approximately three items per order, with a mean quantity of 2.95 and a median of 3. Unit prices vary considerably, ranging from 11.39 to 699.93, with an average of 356.41. The average shopping cart contains approximately five items, while total order values range from 11.39 to 3456.40, with an average of 1053.97 and a median of 823.62. The higher mean compared with the median for TotalPrice suggests the presence of higher-value transactions that increase the average order value.

PRODUCT REVENUE ANALYSIS

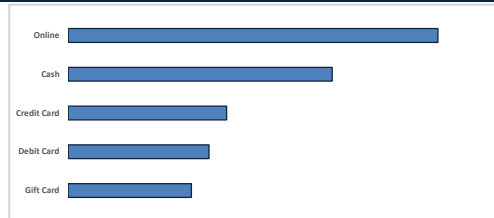
Product	Sum of TotalPrice
Chair	195,620.11
Printer	195,612.61
Laptop	192,126.56
Tablet	186,568.95
Monitor	175,651.41
Desk	167,459.93
Phone	151,722.39



Product revenue varies across the dataset, with Chairs generating the highest total revenue (195,620.11), closely followed by Printers (195,612.61) and Laptops (192,126.56). Tablets (186,568.95) and Monitors (175,651.41) also contribute substantial revenue, while Desks (167,459.93) and Phones (151,722.39) generate comparatively lower revenue, with Phones recording the lowest total. Overall, revenue is fairly well distributed across the product categories, although Chairs and Printers emerge as the strongest-performing products.

PAYMENT METHOD ANALYSIS

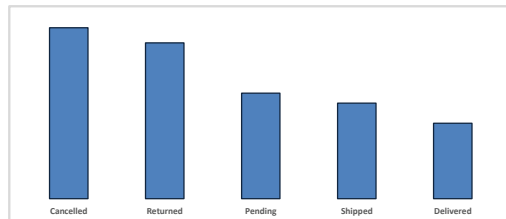
PaymentMethod	Count of OrderID
Gift Card	19.17%
Debit Card	19.33%
Credit Card	19.50%
Cash	20.50%
Online	21.50%



The distribution of payment methods is relatively balanced, with Online payments accounting for the highest proportion of transactions (21.50%), followed by Cash (20.50%) and Credit Card (19.50%). Debit Card and Gift Card payments represent slightly smaller proportions, at 19.33% and 19.17%, respectively. Overall, the payment methods are fairly evenly distributed, indicating that customers utilise a diverse range of payment options rather than relying on a single preferred method.

ORDER STATUS DISTRIBUTION (%)

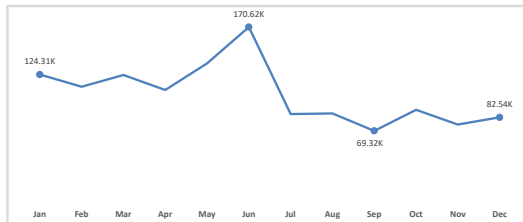
OrderStatus	Count of OrderID
Cancelled	20.83%
Returned	20.58%
Pending	19.75%
Shipped	19.58%
Delivered	19.25%



The distribution of order statuses is relatively balanced, with each status accounting for approximately 20% of total orders. Cancelled orders represent the highest proportion (20.83%), while delivered orders account for the lowest (19.25%), indicating that no single order status dominates the dataset.

MONTHLY TREND

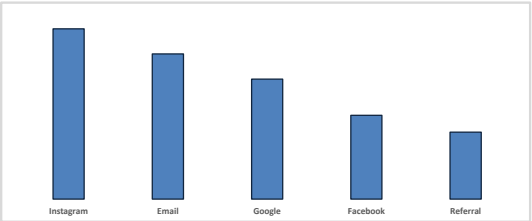
OrderStatus	Sum of TotalPrice
Jan	124313.23
Feb	112344.78
Mar	123840.93
Apr	109186.05
May	135142.59
Jun	170616.13
Jul	85784.64
Aug	86343.21
Sep	69321.65
Oct	89834.82
Nov	75493.43
Dec	82540.5



Monthly sales exhibited notable fluctuations throughout the year. Sales started at a relatively high level of 124,313.23 in January, experienced moderate variations during the first half of the year, and reached a peak of 170,616.13 in June. A sharp decline followed in July, with sales falling to their lowest point of 69,321.65 in September. Sales then recovered modestly during the final quarter, ending the year at 82,540.50 in December. Overall, the data indicate significant month-to-month variability in sales performance.

REFERRAL SOURCE

ReferralSou	Count of OrderID
Instagram	21.58%
Email	20.83%
Google	20.08%
Facebook	19.00%
Referral	18.50%



The distribution of referral sources is relatively balanced, with Instagram contributing the highest proportion of orders (21.58%), followed by Email (20.83%) and Google (20.08%). Facebook and Referral account for smaller proportions, at 19.00% and 18.50%, respectively. Overall, the referral sources are fairly evenly distributed, indicating that customer acquisition is supported by a diverse range of channels rather than a single dominant source.

CONCLUSION

The exploratory data analysis revealed several important patterns within the dataset. Product revenue was relatively well distributed across categories, with Chairs and Printers generating the highest revenue. Customer acquisition and payment methods were evenly distributed, indicating a diverse customer base. Monthly sales exhibited noticeable fluctuations throughout the year, while order statuses remained relatively balanced across categories. Overall, the analysis demonstrates how data visualisation and descriptive statistics can be used to support business decision-making and identify meaningful trends.